

Chandpur Science and Technology University

Lab-Report

Course Code: ICT 3106

Titles: Computer Network Sessional

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Lab 1: Create a Simple Network with a Switch and 2–3 Computers and a Router

Objective

The goal of this lab is to design and configure a basic local area network (LAN) consisting of a switch connected to multiple computers and a router for external connectivity using Cisco Packet Tracer.

Equipment Used

- 3 PCs (end devices)
- 1 Switch (PT-Switch)
- 1 Router (PT-Router)
- Copper straight-through cables

Procedure

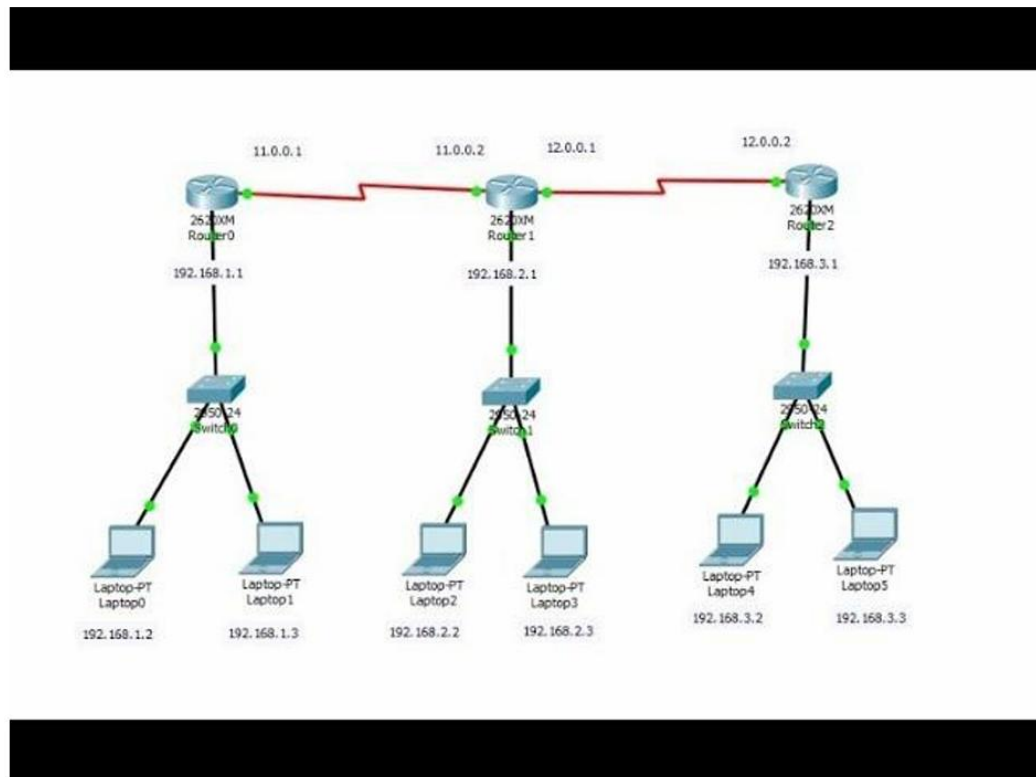
1. Launch Cisco Packet Tracer on your computer.
2. Add the devices to the workspace: Select 3 PCs from the End Devices category, 1 PT-Switch from the Switches category, and 1 PT-Router from the Routers category.
3. Connect the devices:
 - Use copper straight-through cables to connect each PC's FastEthernet0 port to available ports on the switch (e.g., FastEthernet0/1, 0/2, 0/3).
 - Connect the switch's FastEthernet0/4 port to the router's FastEthernet0/0 port using a copper straight-through cable.
4. Configure IP addresses on the PCs:
 - For each PC, go to Desktop > IP Configuration.
 - Assign IPv4 addresses in the 192.168.1.0/24 network (e.g., PC0: 192.168.1.2, PC1: 192.168.1.3, PC2: 192.168.1.4), Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1.
5. Configure the router:
 - Click on the router, go to CLI.
 - Enter privileged mode: enable
 - Enter configuration mode: configure

terminal

- o Configure the interface: interface FastEthernet0/0, ip address 192.168.1.1 255.255.255.0, no shutdown
- o Exit and save: end, write memory

6. Verify connectivity by sending pings between PCs (e.g., from PC0: ping 192.168.1.3).

Diagram



Verification

- Use the Command Prompt on each PC to run ipconfig and confirm IP assignments.
- Ping between PCs and from PCs to the router's IP to ensure successful replies.

Conclusion

This lab demonstrates the setup of a simple star-like LAN with a switch and router, enabling basic communication among devices.

Lab 2: Create a BUS Network with 3 Routers, Also a Cycle of 4 Routers

Objective

To implement a bus topology using 3 routers and a ring (cycle) topology using 4 routers in Cisco Packet Tracer.

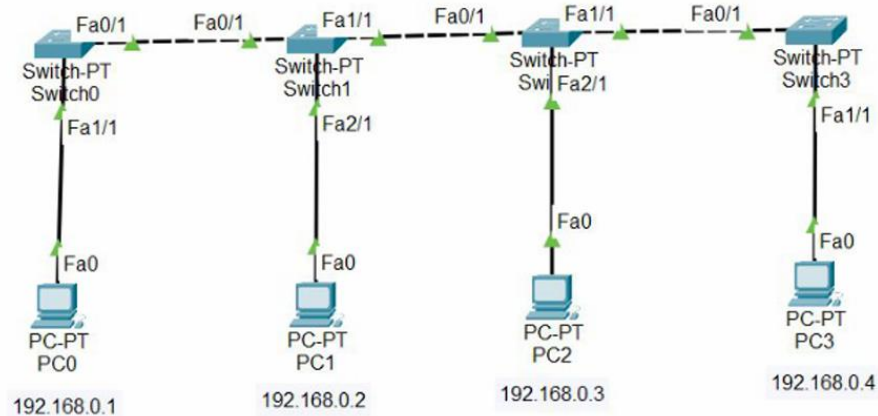
Equipment Used

● For Bus: 3 Routers (PT-Router), Serial cables (DCE/DTE) ● For Ring: 4 Routers (PT-Router), Serial cables (DCE/DTE)

Procedure for Bus Topology

1. Open Cisco Packet Tracer.
2. Add 3 PT-Routers to the workspace.
3. Connect them in a linear bus fashion:
 - Use serial DCE cables: Router0 Serial2/0 (DCE end) to Router1 Serial2/0 (DTE end).
 - Router1 Serial3/0 (DCE) to Router2 Serial2/0 (DTE).
4. Assign IP addresses (e.g., Router0 Serial: 192.168.0.1/30, Router1 Serial to R0: 192.168.0.2/30, Serial to R2: 192.168.0.5/30, etc.).
5. Configure clock rates on DCE ends (e.g., clock rate 64000).

Diagram for Bus Topology



Procedure for Ring Topology

1. Add 4 PT-Routers.
2. Connect in a cycle:
 - o Router0 Serial2/0 (DCE) to Router1 Serial2/0 (DTE).
 - o Router1 Serial3/0 (DCE) to Router2 Serial2/0 (DTE).
 - o Router2 Serial3/0 (DCE) to Router3 Serial2/0 (DTE).
 - o Router3 Serial3/0 (DCE) to Router0 Serial3/0 (DTE).
3. Assign IPs in point-to-point subnets (e.g., /30 masks).
4. Set clock rates on DCE interfaces.

Verification

- Ping across routers (e.g., from Router0 to Router2).
- Use show ip interface brief to check configurations.

Conclusion

This lab illustrates linear (bus) and circular (ring) interconnections of routers, highlighting data flow in shared and looped paths.

Lab 3: Implement FLSM for the Setup on Lab-2

Objective

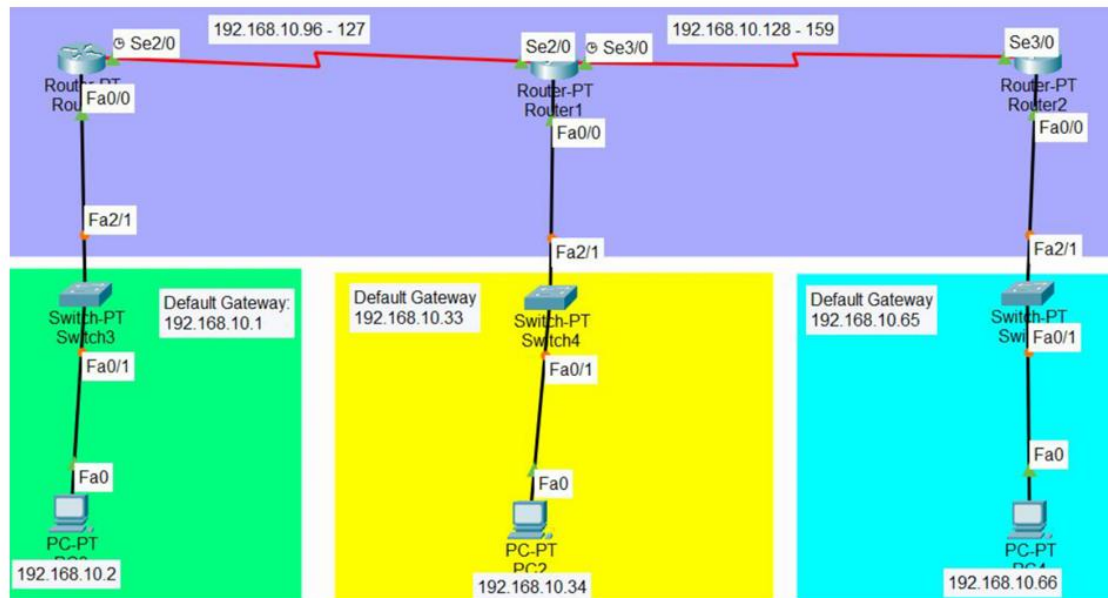
To apply Fixed Length Subnet Mask (FLSM) subnetting to the bus and ring topologies from Lab 2, enabling routed communication. Equipment Used

- Same as Lab 2, plus 1 PC per router for testing (3 for bus, 4 for ring), 1 Switch per router.

Procedure

1. Extend Lab 2 topologies: Add a switch and PC to each router's FastEthernet0/0.
2. Use a Class C network (e.g., 192.168.10.0/24) and subnet into fixed /27 subnets (32 IPs each).
3. Assign subnets: ○ For bus (3 subnets): 192.168.10.0/27, 192.168.10.32/27, 192.168.10.64/27. ○ PCs: Assign IPs from respective subnets (e.g., PC0: 192.168.10.2/27, Gateway: Router0 192.168.10.1/27).
4. Configure router interfaces and static routes (e.g., on Router0: ip route 192.168.10.32 255.255.255.224 192.168.10.98).
5. Repeat similar subnetting for the ring with 4 routers, using additional subnets like 192.168.10.96/27.

Diagram



Verification

- Ping between PCs on different subnets (e.g., PC0 to PC1: ping 192.168.10.34).
- Check routing tables with show ip route.

Conclusion

FLSM divides the network into equal-sized subnets, suitable for uniform host requirements in bus and ring setups.

Lab 4: Implement VLSM with BUS and Ring Topology with 3 and 4 Routers, Also Star Topology

Objective

To implement Variable Length Subnet Mask (VLSM) for efficient IP allocation in bus, ring, and star topologies.

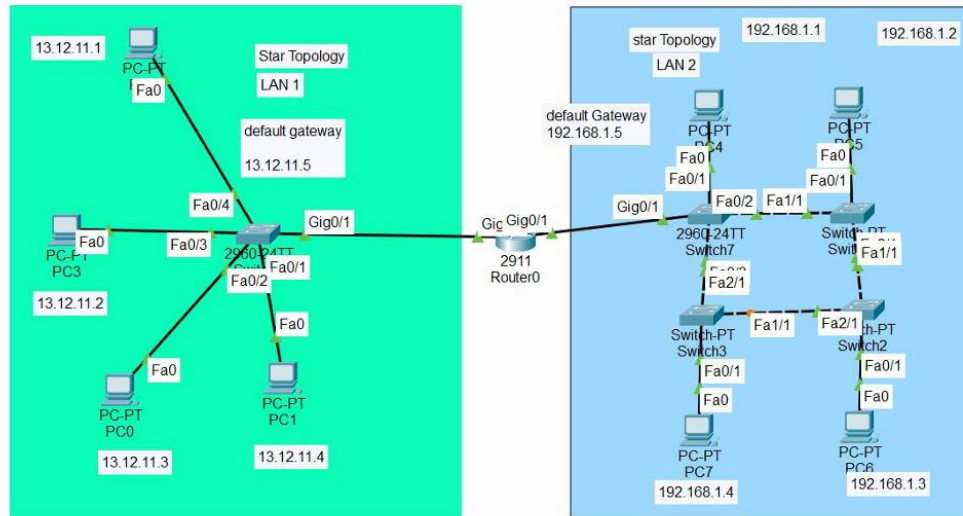
Equipment Used

- Similar to Labs 2-3: 3-4 Routers, Switches, PCs, Cables.
- For Star: Additional central router or switch connecting others.

Procedure

1. Use base topologies from Lab 2.
2. For VLSM, start with 192.168.10.0/24 and subnet variably (e.g., /26 for larger LANs, /30 for links).
 - Bus (3 routers): LANs /27 (32 hosts), links /30 (2 hosts).
 - Assign: PC LANs (e.g., 192.168.10.0/26 for first), serial links 192.168.10.100/30.
3. For Ring (4 routers): Similar, with varying subnet sizes based on needs.
4. For Star: Central router connected to 3-4 peripheral routers (star arms).
 - Apply VLSM: Larger subnet for central, smaller for spokes.
5. Configure interfaces and static routes (e.g., Router0: ip route 192.168.10.64 255.255.255.224 192.168.10.102).

Diagram



Verification

- Ping across varying subnets.
- Verify with show ip route and PDU simulations.

Conclusion

VLSM optimizes IP usage by allowing variable subnet sizes, ideal for diverse topologies like bus, ring, and star.

Lab 5: CLI Command for Above Lab

Objective

To configure and verify the networks from previous labs using Command Line Interface (CLI) commands in Cisco Packet Tracer.

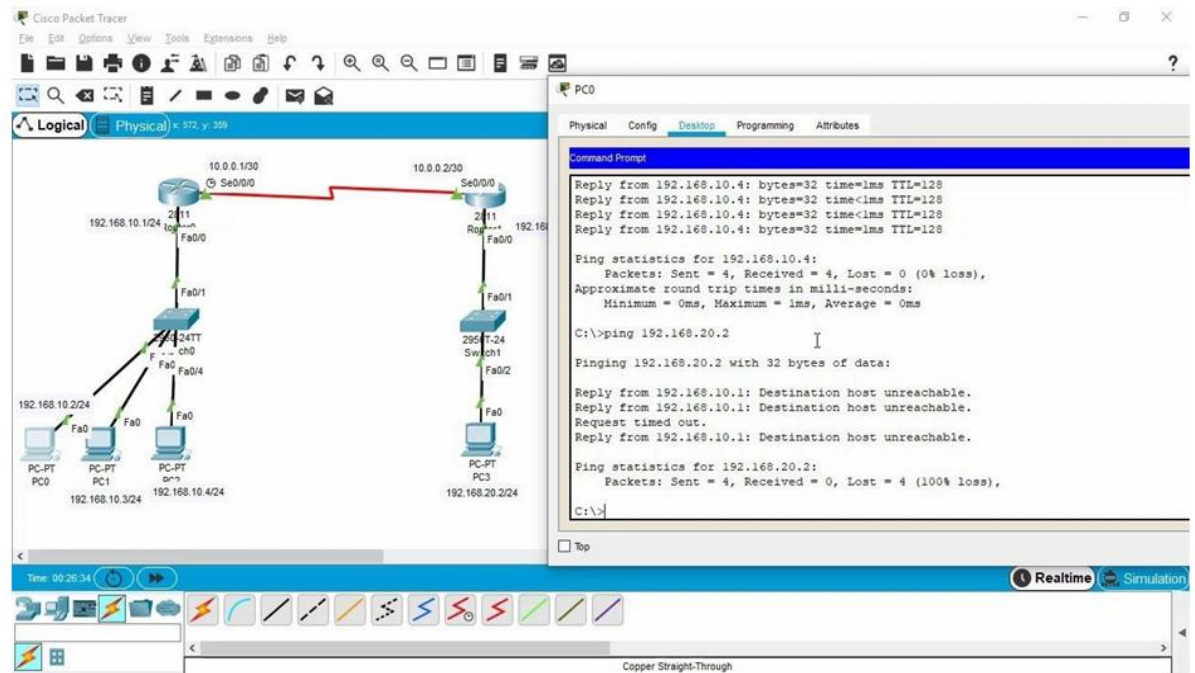
Equipment Used

- Routers from previous labs.

Procedure

1. Access router CLI: Click router > CLI tab.
2. Basic configuration: ○ enable (privileged mode) ○ configure terminal (global config) ○ Interface config: interface FastEthernet0/0, ip address 192.168.10.1 255.255.255.192, no shutdown
3. Static routes (from Labs 3-4): e.g., ip route 192.168.10.64 255.255.255.224 192.168.10.102
4. Verification commands: ○ show ip interface brief ○ show running-config ○ show ip route ○ ping
5. Save: write memory

Diagram



Verification

- Run show commands to confirm configs.
- Test connectivity with ping.

Conclusion

CLI provides precise control for configuring routers in various topologies and subnetting schemes.